

CREATIVE INSTITUTE DATA SCIENCE

Data Cleaning

Practical -2 method of missing value using median & mean

Note- mean & median use for numerical value
mode use for categorical value

first import the essential library

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

then read the data set

```
df=pd.read_csv('train.csv')
df.shape
```

for all columns

```
pd.set_option('display.max_columns',None)
pd.set_option('display.max_rows',None)
```

```
df.head(6)
```

describe the dataset

```
df.info()
```

find any null value?

```
df.isnull()
```

find total null values

```
df.isnull().sum()
```

find the percentage of null values

```
null_var=df.isnull().sum()/df.shape[0]*100
null_var
```

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Data Cleaning

find the column name of whose null values is greater than 20%

```
drop_columns=null_var[null_var>20].keys()  
drop_columns
```

drop the columns whose value is greater than 20%

```
df2_drop_columns=df.drop(columns=drop_columns)  
df2_drop_columns
```

find the shape of new dataframe

```
df2_drop_columns.shape
```

find the only numeric value

```
df3=df2_drop_columns.select_dtypes(include=['int64','float64'])  
df3.head()
```

there is 38 columns where value is present.

Now visualize is there any missing value in new dataset using heatmap

```
sns.heatmap(df3.isnull())
```

it shows there is missing value

find out null value in any row

```
df3[df3.isnull().any(axis=1)]
```

now calculate total missing value

```
df3.isnull().sum()
```

now find out columns name where missing value

```
col_name=[i for i in df3.columns if df3[i].isnull().sum()>0]  
col_name
```

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now we have 3 variable where is missing value

now we need to fill the missing value

we have 3 parameter for filling missing value (mean, median, mode)

for filling value we need to check data distribution using distplot

```
plt.figure(figsize=(10,10))
sns.set()
for i, j in enumerate(col_name):
    plt.subplot(2,2,i+1)
    sns.distplot(df3[j],bins=20,kde_kws={'linewidth':5,'color':'red'})
```

data visualization shows that our data is normal distributed..

when bell curve data appear we can call that bell curve shape to normal distribution or gaussian distribution

when same mirror image appear we should check that data is normally distributed ..

when data is normal distributed we use (mean, median, mode)

general we use mean for missing value

Note: when missing value is random we use mean & median only

Now we need to fill missing value using fillna() function using mean

and check if there any null value

```
df4=df3.fillna(df3.mean())
df4.isnull().sum().sum()
```

now we need to compare distribution of original dataset and new dataset

```
plt.figure(figsize=(10,10))
sns.set()
for i, j in enumerate(col_name):
    plt.subplot(2,2,i+1)
    sns.distplot(df3[j],bins=20,kde_kws={'linewidth':8,'color':'red'},label='original')

    sns.distplot(df4[j],bins=20,kde_kws={'linewidth':5,'color':'green'},label='new')
plt.legend()
```

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you can clearly see that original dataset and new dataset distribution is.

You can visualize there is no difference in 2,3 graph and difference in lotfrontage graph because of mean fill value

Now check using median value to fill missing value in dataset

```
df5=df3.fillna(df3.median())  
df5.isnull().sum().sum()
```

now check distribution using distplot of mean median

```
plt.figure(figsize=(10,10))  
sns.set()  
for i, j in enumerate(col_name):  
    plt.subplot(2,2,i+1)  
    sns.distplot(df3[j],bins=20,hist=False,kde_kws={'linewidth':8,'color':'red'},label='original')  
    sns.distplot(df4[j],bins=20,hist=False,kde_kws={'linewidth':5,'color':'green'},label='mean')  
    sns.distplot(df5[j],bins=20,hist=False,kde_kws={'linewidth':3,'color':'k'},label='median')  
plt.legend()
```

you can clearly see that there is slightly distribution on the both chart.

When dataset has outlier there is difference in mean & median

Now for checking the fill value of mean & median

We need to concate all the data

```
df_concat=pd.concat([df3[col_name],df4[col_name],df5[col_name]],axis=1)  
df_concat[df_concat.isnull().any(axis=1)]
```

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Now find out is there any missing value in present in the new dataset

```
df3.isnull().sum()
```

now we clearly find out that there is only 3 columns whose value is missing.

Now find out is there any null value present in this row for 3 columns

```
l=['LotFrontage','MasVnrArea','GarageYrBlt']  
df3[l][df3[l].isnull().any(axis=1)]
```

now we need to fill value based on

now visualize is there any missing value in new dataset

```
plt.figure(figsize=(25,25))  
sns.heatmap(df3.isnull())
```

you can clearly visualize there is 3 columns in missing value

now we need to find out missing value in row

```
df3[df3.isnull().any(axis=1)]
```